# Problem Set 2

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**Problem 1. (BdMO 2019 Higher Secondary P2)** Prove that, for any positive real numbers $a, b, c$

$$\frac{a}{bc} + \frac{b}{ca} + \frac{c}{ab} \geq \frac{2}{a} + \frac{2}{b} - \frac{2}{c}$$

**Problem 2. (Baltic Way 2021 P16)** Show that no non-zero integers $a$, $b$, $x$, $y$ satisfy

$$\begin{cases} ax - by = 16 \\ ay + bx = 1 \end{cases}$$

**Problem 3. (Al-Khwarizmi IJMO 2024 P5)** At a party, every guest is a friend of exactly fourteen other guests (not including him or her). Every two friends have exactly six other attending friends in common, whereas every pair of non-friends has only two friends in common. How many guests are at the party?

**Problem 4. (Simurgh 2019 P2)** Let $ABC$ be a triangle with $AB = AC$. Let point $Q$ be on plane such that $AQ \parallel BC$ and $AQ = AB$. Now let the $P$ be the foot of perpendicular from $Q$ to $BC$. Show that the circle with diameter $PQ$ is tangent to the circumcircle of triangle $ABC$.

**Problem 5. (ALMO 2025 Junior P1)** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(2x + y)) + f(x) = 2x + f(x + y)$$

for any $x, y \in \mathbb{R}$.

**Problem 6. (Dutch MO 2025 P2)** Find all pairs $(n, p)$ such that

$$n! - (n - 1)n = 4p^2$$

where $n$ is a positive integer and $p$ is a prime number.

**Problem 7. (India IMOTC 2024 P1)** A sleeping rabbit lies in the interior of a convex 2024-gon. A hunter picks three vertices of the polygon and he lays a trap which covers the interior and the boundary of the triangular region determined by them. Determine the minimum number of times he needs to do this to guarantee that the rabbit will be trapped.

**Problem 8. (Silk Road 2024 P2)** Let $ABCD$ be a non-isosceles trapezoid with $AB \parallel CD$. A circle through $A$ and $B$ meets $AD, BC$ at $E, F$. The segments $AF, BE$ meet at $G$. The circumcircles of $\triangle ADG$ and $\triangle BCG$ meet at $H$. Show that if $GD = GC$, $H$ is the orthocenter of $\triangle ABG$.